

Semantic Field Overlap

KWTA applied to word activation maps.

Store a dataset of 1.8M word associations (e.g. "ship"→"sail") in hetero-associative memory. Retrieve activation maps for two input tokens and compute their overlap using kWTA.

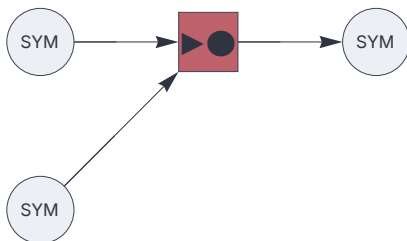
```
<< "Circuits.m"
```

Hetero-associative memory circuit

```
hyperparameters = {5000,10};
```

```
dataflow = {
    input[codec] [1] [],
    input[codec] [2] [],
    associator[] [11] [2, 1],
    output[codec] [] [11],
    _ → hyperparameters
};
```

```
c = Circuit[dataflow]; c[Graph, ImageSize→ 220]
```

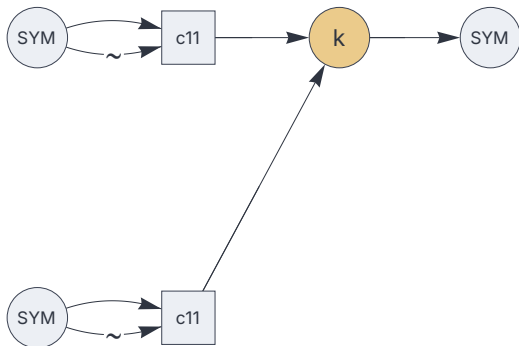


Testing circuit

Embedding the memory circuit above.

```
testcircuit = {
  input[codec] [11] [],
  input[codec] [12] [],
  circuit[c][21][11, 11"~"], (* "~" clears data *)
  circuit[c][22][12, 12"~"],
  kwt[a][31][{21,22}],
  output[codec] [] [31],
  _ → hyperparameters,
  "multiplex"→"10" (* One empty input to flush the delay. *)
};
```

```
test = Circuit[testcircuit]; test[Graph, ImageSize→ 280]
```



Import dataset

```
data = Import[ NotebookDirectory[] <> "/data/word-associations.csv.zip",
  "word-associations.csv"];
```

```
{hobbyists ↔ collectors, ballpark ↔ diamond, sighting ↔ appearance,
superfine ↔ wool, sensorimotor ↔ control, sneaking ↔ tiptoe,
microclimate ↔ ecosystem, patternless ↔ unordered,
totality ↔ perfectness, squalidly ↔ sordid}
```

```
1 888 505
```

Train

```
c @@@ data;
```

Test

Retrieve and decode semantic fields:

```
c[ "ship", Null]
```

```
Sequence[{harbor, mast, stern, bow, deck, helm, nautical,  
ocean, marine, port, starboard, navigation, maritime, boat,  
vessel, crew, sail, voyage, launch, sea, anchor, fleet, seas,  
captain, sailing, compass, cargo, dock, hull, shipyard, keel}]
```

```
c["boat", Null]
```

```
Sequence[{trip, harbor, mast, stern, bow, deck, nautical, fishing, ocean,  
maritime, river, craft, water, vessel, ferry, crew, passenger, sail,  
ride, voyage, sea, anchor, owner, channel, pier, motor, speed, navigate,  
ramp, seas, wave, engine, captain, sailing, raft, tour, rudder, buoyancy,  
lake, yacht, dock, knot, row, canoe, hull, mooring, builder, paddle,  
stumped, marina, buoy, wharf, keel, oar, dinghy, regatta, lifeboat}]
```

Compute kWTA, using the testing circuit.

```
test["ship", "boat"]
```

```
Sequence[{harbor, mast, stern, bow, deck, nautical, ocean, maritime, vessel,  
crew, sail, voyage, sea, anchor, seas, captain, sailing, dock, hull, keel}]
```